INCH-POUND

MIL-R-49465/3A
6 January 1992
SUPERSEDING
MIL-R-49465/3
18 April 1988

MILITARY SPECIFICATION

RESISTORS, FIXED, METAL ELEMENT (POWER TYPE),

STYLE RLV21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RLV21 metal element, power type, fixed resistors.

1.2 Military Part or Identifying Number (PIN). The resistors covered by this specification shall be identified by a military PIN which shall consist of the basic number of this specification and a coded number. The military PIN shall be in the following form:



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-RS, Fort Monmouth, NJ 07703-5302, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5905

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-49465 - Resistors, Fixed, Metal Element (Power Type), (Very Low Resistance Values), General Specification For.

MIL-R-39032 - Resistors, Packaging of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-49465, and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Characteristic. This style resistor is available in characteristic L as described in MIL-R-49465.

3.4 Tolerance. This style resistor is available in tolerances J and K (±5 percent and ±10 percent respectively).

3.5 Terminal lead types. This style resistor is a four-leaded device with solderable terminals (see figure 1).

3.6 Weight. Weight shall be .013 lbs (5.90 grams) as shown in figure 1.

3.7 Power rating. The power rating shall be 5.0 watts at 25°C as shown in table I.

3.8 Maximum overload current. The maximum overload current shall be as specified in table I.

3.9 Resistance value range. The minimum and maximum resistance values shall be as specified in table I.

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3.10 Temperature coefficient. The temperature coefficient of resistance shall be as specified in table II.

3.11 Marking. The resistors shall be marked in accordance with MIL-R-49465.

3.12 Potting compound characteristics. Due to the nature of this design, if units are potted in a ceramic "boat" a natural shrinkage of the potting may occur prior to or following some tests. This will not be considered a defect as long as it does not impair the electrical or environmental characteristics. The internal circuitry shall not be exposed. Cracks in the boat itself are not allowed.

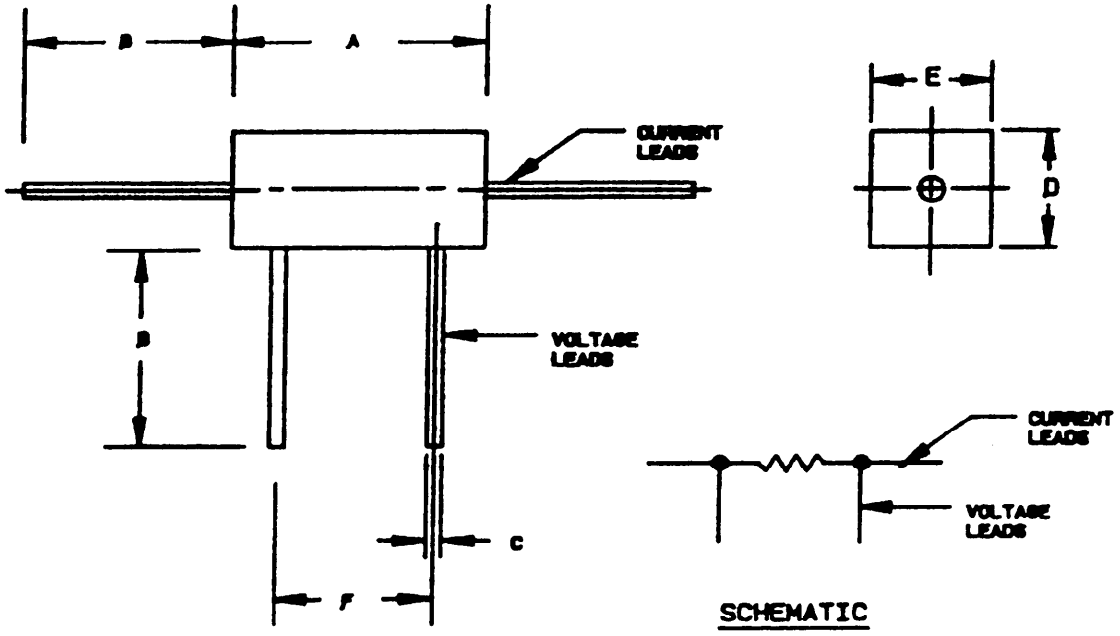
TABLE I. Electrical characteristics.

Style	Rated wattage at 25°C (watts)	Minimum resistance (ohms)	Maximum resistance (ohms)	Maximum overload current (amperes)
RLV21	5.0	0.010	0.10	32.0

TABLE II. Temperature coefficient of resistance.

Style	.01 ohm to .0249 ohm	.025 ohm to .0499 ohm	.05 ohm to .0749 ohm	.075 ohm to .099 ohm	0.1 ohm and above
	(ppm/°C)	(ppm/°C)	(ppm/°C)	(ppm/°C)	(ppm/°C)
RLV21	±100	±100	±100	±100	±100

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Style	Maximum weight lbs(grams)	Dimensions					
		A ±.031 (±.787)	B min.	C ±.005 (±.130)	D ±.031 (±.787)	E ±.031 (±.787)	F ±.060 (±.152)
RLV21	.013 (5.90)	.875 (22.23)	1.00 (25.40)	.038 (0.90)	.343 (8.71)	.375 (9.53)	.562 (14.27)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Lead length shall be 1 inch minimum.
5. For resistance measurements the point of contact to the leads shall be .375 (9.52 mm) ±.062 (1.57 mm) from the body.
6. The picturization of the style above is given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope and do not alter the functional aspects of the device, are acceptable.

FIGURE 1. Resistors, fixed, metal element.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-49465.

4.2 Visual and mechanical inspection. Visual and mechanical inspection shall be in accordance with MIL-R-49465.

4.3 Terminal strength. In accordance with MIL-R-49465 and the applied force shall be five pounds.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

6.1 Notes. The notes specified in MIL-R-49465 are applicable to this specification.

6.2 Ordering data. Ordering data shall be as specified in MIL-R-49465.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 99
DLA - ES

User activities:
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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